MILITARY SPECIFICATION

NAVIGATIONAL SET, TACAN AN/ARN-105

This specification has been approved by the Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - The equipment covered by this specification shall, when installed in aircraft, provide bearing, distance and tone identity information to indicate the location of a complementary surface station with respect to the aircraft. The equipment shall also provide, as a discrete auxiliary function, information to indicate the link distance with respect to a similarly equipped aircraft. This equipment is the airborne portion of the short range air navigation system known as TACAN (Tactical Air Navigation).

1.2 Classification - The equipment covered by this specification shall consist of the following items:

Type Designation	Applicable Paragraph
RT-1030/ARN-105	3.5.1
C-2010/ARN-52(V) or	3.5.2
C-2010A/ARN-52(V) or	3.5.3
C-3928(P)/ARN-52(V) or	3.5.3.7
C-7893(P)/ARN-52(V)	3.5.3.8
C-7712/ARN-52(V)	3.5.3.9
MT-2091/ARN-65	3.5.4,

1.3 Associated Equipment - This equipment shall operate with the associated equipment listed in 6.7.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

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SPECIFICATIONS

Military

MIL-C-172	Cases, Bases, Mounting; and Mounts, Vibration (for use with Electronic Equipment in Aircraft)
MIL-W-5088	Wiring; Aircraft, installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements Aircraft Equipment
MIL-P-7788	Plate, Plastic, Lighting
MIL-M-7793	Meter, Time Totalizing
MIL-I-9229	Indicator, Course ID-387/ARN
MIL-E-17555	Electronic, and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures, Preproduction and Inspection for Aircraft Electronic Equipment, format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment.
MIL-E-19600	Electronic Modules, General Specifications for
MIL-I-22075	Indicator, Bearing-Distance-Heading: ID-663/U

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MIL-C-5824	Course Indicator ID-250A/ARN
MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted
MIL-I-18955	Indicator, Range ID-388/ARN
MIL-A-25708	Antenna AT-741/A
MIL-I-23366	Horizontal Situation Indicator ID-1013/A
MIL-S-25879	Switch, Radio Frequency Transmission Line SA-521/A
MIL-I-27848	Horizontal Situation Indicator AQU-4/A
MIL-I-25999	Indicator, Bearing, Distance Heading ID-526A
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-H-26689	Horizontal Situation Indicator AQU-2/A

Naval Air Systems Command

AR-5	Microelectronic Devices Used in Avionics Equipment, Procedures for Selection and Approval of.
AR-10	Maintainability of Avionics Equipment and Systems, General Requirements for.
AR-34	Failure Classification for Reliability Testing, General Requirements for

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STANDARDS

Military

MIL-STD-287	Test Signals for Electronic Tactical Air Navigation Equipment (TACAN)
MIL-STD-291	Standard Tactical Air Navigation (TACAN) Signal
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of.
MIL-STD-794	Parts & Equipment, Procedures for Packaging and Packing of.
MIL-STD-785	Reliability Program for System and Equipment Development and Production.
MIL-STD-781	Reliability Tests, Exponential Distribution.
MIL-STD-810	Environmental Test Methods

2.1.1 Availability of Documents - When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publications and Forms Center, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

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(1) Microelectronic Technology shall be considered and microelectronic items shall conform to requirements specified herein.

(2) Other parts and materials requirements shall conform to Specification MIL-E-5400.

(3) Nonrepairable subassemblies shall be used in accordance with Specification AR-10 and as outlined in Specification MIL-E-5400.

(4) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials (including electron tubes, transistors and diodes) other than microelectronic devices shall be obtained as outlined in Specification MIL-E-5400. Microelectronic devices shall be approved as outlined in AR-5.

3.2.2 Microelectronic Modular Assemblies - When used, Microelectronic Modular Assemblies shall meet the requirements of Specification AR-5. Conformal coatings, encapsulant, embedments or potting materials used with modular assemblies containing integrated circuits and discrete parts shall be easily removable without damage to the assembly.

3.2.3 Modules - The electronic portions of the equipment shall be functionally modularized in accordance with Specification AR-10.

3.2.4 Motors - All motors shall have sealed bearings . that do not require lubrication or other service for a minimum of 5000 hours of equipment operation.

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

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3.3.1 Total Weight - The total weight of the equipment excluding cables, shall be a minimum consistent with good design and shall not exceed 48 pounds.

3.3.2 Reliability - The Contractor shall conduct a reliability program using MIL-STD-785 as a guide. On a reorder from a supplier who has previously produced the equipment, the program previously used may be continued unless otherwise indicated in the contract or order.

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 250 hours without the necessity for readjustment of any controls which arc inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Equipment Reliability in Mean-Time-Between Failures (MTBF) - The equipment shall have a specified mean (operating) time between failures (MTBF) of 175 hours when tested and accepted as outlined under the requirements of 4.4.3.

3.3.2.4 Time Totalizing Meter - The following units shall contain time totalizing meters in accordance with Specification MIL-M-7793:

Unit	Type of Meter
RT-1030/ARN-105	MS-28053-5

3.3.3 Cabling and Connections:

3.3.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

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3.3.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5085. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.3.4 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all panels must be approved by the procuring activity prior to preproduction testing.

3.3.5 Interchangeability - The equipment shall meet the interchangeability requirements of Specification MIL-E-5400.

3.3.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.3.7 Provisions for Maintainability - The Maintainability Program, built-in test features, construction and packaging, provisions for test points, and other maintainability parameters shall be as specified in Specification AR-10.

3.3.8 Nomenclature, Nameplates and Identification Marking - Nomenclature and serial number assignment, nameplate approval and identification marking shall be in accordance with Specification MIL-N-18307.

3.3.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests:

Temperature	Room ambient (25°C ± 5°C)
Altitude	Normal Ground

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Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input Power Voltage	115 ± 1.0 VAC and 27.5 ± 0.5 VDC

3.3.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental. service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 1 equipment except as specified herein:

3.3.10.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curve II of MIL-E-5400 and Figure 6 of this document.

3.3.10.2 Temperature - The equipment shall operate satisfactorily when subjected to the temperature requirements of Specification MIL-E-5400 Class 1 equipment, except that the requirement for intermittent use shall be 60 minutes.

3.3.10.3 Altitude - The maximum altitude shall be 70, 000 feet.

3.3.10.4 Cooling - Cooling provisions shall be such as to maintain parts within their permissible operating temperature under all service conditions or reasonable combinations of service conditions.

3.3.11 Warm-Up Time - The time required for the equipment to warm up prior to operation shall be kept to a minimum and shall not exceed 90 seconds under standard conditions and 3 minutes at extreme service conditions.

3.3.12 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts.

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(1) AC Power (Single Phase) 115V, Category B, 310VA

(2) DC Power 28V, Category B, 2.2 Amps

3.3.12.1 Degraded Performance - Degraded performance will be permitted for transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the equipment.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service and input power conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3. 4.1 Operation - A working assembly of the equipment when properly interconnected shall transmit to and receive from a similarly equipped complementary aircraft or a complementary navigation beacon a combination of pulsed signals from which it shall derive and process the information necessary to:

(1) Continuously display the bearing to the beacon in the REC (receive) and T/R (transmit-receive)

(2) Indicate the deviation of the aircraft from the selected course or beacon radial in the REC (receive) and T/R (transmit-receive) modes.

(3) Indicate the distance to the beacon in the T/R (transmit-receive) mode.

(4) Indicate the distance to the complementary aircraft in the A/A (air-to-air) mode.

(5) Transpond to the complementary aircraft's interrogation in the A/A (air-to-air) mode.

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(6) Identify, by aural identification, the beacon to which the equipment is channeled in the REC (receive) and T/R (transmit-receive) modes.

(7) Automatically select, for satisfactory performance, one of two associated antennas in REC (receive), T/R (transmit-receive) and A/'A (air-to-air) modes.

Furthermore, the equipment, independently of the signals received from the beacon or complementary aircraft, shall provide a blanking pulse output during the time of its own transmission in the T/R (transmit-receive) and A/A (air-to-air) modes which will cause blanking of other equipment operating in the same frequency band.

3.4.2 Characteristics of TACAN Signals - This equipment shall operate with pulse code and pulse amplitude type signals of a surface or air facility meeting the requirements of MIL-STD-291.

3.4.3 Frequency Characteristics - One hudred twenty-six (126) crystal controlled frequency channels shall be provided at one megahertz spacing. In the T/R mode, channels 1 through 63 shall pair receiver frequencies 962 through 1024 megahertz with transmitter frequencies 1025 through 1087 megahertz and channels 64 through 126 shall pair receiver frequencies 1151 through 1213 megahertz with transmitter frequencies 1088 through 1150 megahertz. In the REC and T/R modes the receiver shall be tuned identically. In the A/A mode, channels 1 through 63 shall pair receiver frequencies 1088 through 1150 megahertz with transmitter frequencies 1025 through 1087 megahertz and channels 64 through 126 shall pair receiver frequencies 1025 through 1087 megahertz with transmitter frequencies 1088 through 1150 megahertz.

3.4.4 Modulation Characteristics - Modulation characteristics shall be as follows when the RF pulse envelope is detected by a linear detector:

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(1) Pulse Top - The instantaneous magnitude of the pulse shall not, at any instant between the point on the leading edge which is 95% of the pulse amplitude and the point on the trailing edge which is 95% of the pulse amplitude, fall below 95% of the pulse amplitude.

(2) Pulse Duration - The pulse duration measured between the 50% amplitude points shall be 3.5 ±0.5 microseconds.

(3) Pulse Coding - In the T/R mode, the pulse code spacing-measured between the 50% amplitude points from leading edge to leading edge shall be 12.0 ± 0.5 microseconds. This requirement shall also apply to the interrogating pulses in the A/A mode. The transmitted replies in the A/A mode shall be single pulses.

3.4.5 Pulse Repetition Rate - When the equipment is interrogating a beacon or another aircraft, the maximum pulse repetition rate shall be 150 pulse pairs per second (pp/s) in the search mode and 30 pp/s in the track mode. The pulse repetition rate shall vary or jitter enough to preclude synchronization of transmitted pulses amongst any of 100 equipments on the same channel for more than one second in one hour. A search repitition rate of 135 pp/s shall be avoided to preclude synchronization with the fine bearing information.

3.4.6 Transmitter Spectrum - In both the search and track conditions, 94% of the transmitted energy shall be contained within a 500 kilohertz band centered on the nominal frequency of the assigned channel. Each lobe of the spectrum shall be of less amplitude than the adjacent lobe nearer the nominal channel frequency.

3.4.7 Transmitter Power Output:

3.4.7.1 Transmitter Power in T/R Mode - When the equipment is interrogating a beacon in the T/R mode, the peak power of each pulse transmitted during the search condition shall be not less than 1.5 kw nor more than 3.0 kw on all channels when the transmitter is terminated in a 50 ohm resistive load impedance.

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3.4.7.1 (continued)
The transmitted output shall not drop below 0.95 kw with a two to one standing wave ratio of any phase angle at the transmitter. The two pulses in an interrogating pulse pair shall not differ in power by more than 0.5 db when the load impedance is 50 ohms.

3.4.7.2 Transmitter Power in A/A Mode - With the equipment operating in the A/A mode at a transmitting rate of 150 pulse pairs per second, the requirements of 3.4.7.1 shall apply. The transmitter power shall recover to within 2 db of the maximum peak power within 200 micro-seconds after any transmission and shall be at least 750 watts when the transmitter rate is 1050 pulses per second.

3.4.7.3 Effect of Transmitter Power on Other Functions - Under no circumstances shall the equipment lock-on to its own transmission. No degradation of any of the other functions of the equipment shall occur as a result of its own transmission.

3.4.8 Traffic Capacity in A/A Mode - With the equipment operating in the A/A mode on a given channel, it shall be pcssible to transpond effectively to each of five interrogating aircraft when their - equipments are each set 63 channels from the given channel and when their signal strengths at the antenna of the equipment specified herein are within 12 db of each other. The equipment specified herein shall interrogate and lock-on to one of the other five, but it shall not be required to determine the one to which it is locked-on. The equipment shall not be damaged when subjected to excessive interrogations.

3.4.9 Sensitivity - The equipment sensitivity shall be as follows:

Distance Measurement	-90 dbm
Bearing Measurements	-80 dbm
Beacon Identification	-90 dbm

3.4.10 Radio Interference

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3..4.10.1 On Frequency Interference - With the equipment tracking on a signal of -80 dbm, the specified performance shall not be affected by the following types and levels of interference on or within 3 megahertz of the desired or image frequency:

Type	Level
(1) 6000 randomly spaced single pulses per second each 3.5 microseconds wide (in REC and T/R modes only.)	-20 dbm
(2) CW (in REC, T/R and A/A modes)	

Type		Level	
		Image	On Freq.
(2.1)	Within 400 KHz of tracked signal		
(2.1.1)	Bearing	-50 dhm	-110 dbm
(2.1.2)	Identity Tone	-35 dbm	- 95 dbm
(2.1.3)	Distance	-35 dbm	- 95 dbm
(2.2)	400 KHz to 3 MHz from tracked signal		
(2.2.1)	Bearing	-25 dbm	- 85 dbm
(2.2.2)	Identity Tone	-15 dbm	- 75 dbm
(2.2.3)	Distance	-15 dbm	- 75 dbm

3.4.10.2 Off Frequency Interference - The noise level at the output of the second detector with no desired signal present shall not be changed by CW signals of the following frequencies and amplitudes at the antenna terminal:

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Frequency	Amplitude
0.1 MHz to 950 MHz	-30 dbm
6.3 MHz (IF)	-10 dbm
1250 MHz to 10,000 MHz	-30 dbm

3.4.10.3 Radiated and Conducted Interference - The equipment shall meet the radiated and conducted interference and susceptibility requirements of MIL-I-6181. The equipment shall comply with these requirements in the REC, T/R and A/A modes of operation, including both the search and track conditions, and also during channel selection. Furthermore, the antenna conducted spurious CW power shall not exceed -47 dbm at frequencies between 950 and 1250 MHz in the REC mode or between pulse pairs in the T/R and A/A modes.

3.4.11 Operational Range - The operational range of the equipment shall be 300 ± 3 nautical miles. The equipment shall be required to perform all functions specified herein with the prescribed accuracies while operating in its respective modes within the operational range.

3.4.12 Direction of Search:

3.4.12.1 Distance Search Direction - When the equipment electrically scans the range in search of a reply signal, the direction of scanning shall be outward in distance in order to minimize erroneous distance data resulting from echoes.

3.4.12.2 Bearing Search Direction - In the absence of a usable bearing signal, the equipment shall cause the bearing indicator to search in the direction which produces successively retrogressive bearing indications. When a usable bearing signal is available at the antenna terminal, the equipment shall cause the bearing indicator to move in the direction which involves the shorter angular coverage between the position at the time of application of the signal and the position of correct bearing display.

3.4.13 Speed of Response:

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3.4.13.1 Distance speed of Response - It shall take no longer than 30 seconds to search, track and display the transponder distance signals when the transponder reply efficiency is 70%.

3.4.13.2 Bearing Speed of Response - It shall take no longer than 8 seconds to search, track and display the proper beacon bearing signal.

3.4.14 Tracking Rate:

3.4.14.1 Distance Tracking Rate - When operating with a transponder or beacon simulator which is only 70% efficient in replying to interrogations, the equipment shall be capable of locking-on and continuously tracking distance replies while moving at speeds up to 3500 knots in either direction along the transponder radial.

3.4.14.2 Bearing Tracking Rate - The equipment shall be capable of locking on—and maintaining track in bearing during a rate of change in either direction of 20 degrees per second.

3.4.15 Memory:

3.4.15.1 Distance Memory - The equipment shall delay 9 to 12 seconds before searching for a lost distance signal which had been present previously for at least 15 seconds.

3.4.15.2 Bearing Memory - The equipment shall delay 3 to 8 seconds before searching for a lost bearing signal which had been present previously for at least 15 seconds.

3.4.16 Accuracy - The accuracies specified below shall be determined using a standard TACAN signal as defined by Specification MIL-STD-287 and as modified herein. These accuracies shall be realized when the output loading of the equipment is consistent with the requirements of 3.5 and the equipment is operated under any of the service conditions or reasonable combination thereof as specified in Specification MIL-E-5400.

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3.4.16.1 Distance Accuraccy - The equipment shall have a distance accuracy of ±0.1 mile ±0.2% of the distance within the range of 0 to 50 miles and ±0.2 mile from 50 miles to maximum range under both static and dynamic tracking conditions when the distance indicator is set to zero at zero miles. This accuracy shall be maintained over a range of signal strength from -90 dbm to 0 dbm. An additional error of 0.1 mile in distance accuracy over any distance range is permitted following extended exposure to salt spray atmosphere.

3.4.16.2 Bearing Accuracy - The equipment shall have a bearing accuracy as shown in the table below under static tracking conditions.

Bearing Accuracy	Signal Strength
± 2°	-80 dbm to -77 dbm
± 1°	-77 dbm to -5 dbm
± 2°	-5 dbm to 0 dbm

The increase in error under dynamic tracking conditions shall not exceed 1° at 5°/sec. The above shall apply at all bearings and within the modulation limits 15% to 55% with modulation indexes not more than 0.30 nor less than 0.07.

3.4.16.2.1 15 Cycle Accuracy - The equipment shall have a bearing accuracy of ±6° when operating with 15 cycle information, only. This requirement shall be applicable over a 15% to 55% total range of modulation with the 15 cycle modulation index of not more than 0.30 nor less than 0.07.

3.4.16.2.2 External Phase Shift - The equipment shall be capable of locking on and maintaining lock with a bearing rate of change of 5° per second when the phase difference between the 15 cycle and 135 cycle components of the modulation envelope external to the equipment is ±15° (at 15 cps.)

3.4.17 Beacon Identification - In the REC and T/R modes the equipment shall provide intelligible and unambiguous aural identification of the selected beacon signal with a 10 db signal-plus-noise ratio at a signal strength of -90 dbm. The beacon shall be identifiable by a Morse coded audio tone of 1350 cycles per second.

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3.4.18 Remote Control - Complete operational remote control of the receiver-transmitter shall be provided throughout the control panel.

3.4.19 Selectivity:

3.4.19.1 Adjacent and other Channels - The equipment shall have at least 60 db rejection to any beacon signals complying to MIL-STD-291 on any-channel other than the channel to which the equipment is tuned. This requirement applies both in the presence or absence of a desired beacon signal on the selected frequency channel. In the presence of a desired signal, the undesired signal shall be considered rejected if all the performance requirements of this specification are met and no bearing errors result from the introduction of the interfering signal. In the absence of a desired signal, the undesired signal shall be considered rejected if no distance or bearing information is obtained.

3.4.19.2 Co-Channel - The equipment shall be capable of rejecting the weaker of two beacon signals received on the same channel when the difference in signal strength is 8 db or more. The performance obtained from the stronger signal shall meet all requirement of this specification except that an additional one degree bearing error is permitted, when only 8 db of difference exists between the two signals. The 8 db difference shall be measured using test signals each having equal modulation indexes, but not exceeding a total of 40% modulation.

3.4.20 Pulse Code Discrimination - The sensitivity of the receiver shall not decrease more than 3 db when receiving a pulse pair spaced ±0.5 microsecond from nominal. At signal levels between -75 dbm and 0 dbm the receiver shall not lock-on in range and bearing to pulse pairs spaced ±4.0 microseconds from nominal. At signal levels below -75 dbm the receiver shall not lock-on in range or bearing to pulse pairs spaced ±4.5 microseconds from nominal.

3.4.21 Channel Selection Time - Channel selection shall be possible under all conditions of 3.4. Selection time for one channel to any other channel shall not exceed 5 seconds under standard conditions and 60 seconds under extreme service conditions after the normal warm-up time.

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3.4.22 System Delay:

3.4.22.1 Normal TACAN System Delay - The equipment shall be so calibrated in the T/R mode as to effect an indication of zero ±0.1 mile when the delay at the antenna terminal between the 50% amplitude points on the respective leading edges of the second RF interrogation pulse of a coded pair and the corresponding second RF reply pulse of a coded pair is 50 microseconds.

3.4.22.2 Air-to-Air System Delay - The equipment shall be so calibrated in the A/A mode as to effect an indication of zero ±0.1 mile when the delay at the antenna terminal between the 50% amplitude points on the respective leading edges of the second RF interrogation pulse of a coded, transmitted pair and the corresponding received RF reply pulse is 50 microseconds. The delay at the antenna terminal between the 50% amplitude points on the respective leading edges of the second RF interrogation pulse of a coded, received pair and the corresponding transmitted RF reply pulse shall be 50 ±0.5 microseconds when the level of the received signal at the antenna terminal is -40 dbm.

3.4.23 Frequency Channels - The receiver-transmitter shall provide 126 crystal controlled channels as specified in MIL-STD-291. To insure reduced spurious radiations, 126 crystals shall be employed, one per channel. The receiver-transmitter shall be automatically electrically tuned to the desired channel by means of the remote channel selector switch on the control unit.

3.4.24 Impedance Levels - The RF input and output shall provide for a common connection to a 52 ohm coaxial line. The receiver and transmitter shall not be damaged under any mismatched conditions.

3.4.25 Antenna Switching - An antenna switching feature shall be incorporated in the equipment for the purpose of selecting one of two associated antennas. The circuit shall be so designed as to accept the first antenna which provides satisfactory signal level. The equipment shall provide an “either-or” voltage for the control of the antenna switch. The antenna switch shall not be a part of the equipment specified herein and shall be a type SA-521/A in accordance with Specification MIL-S-25879 or an approved equivalent. In the absence of a usable signal the antenna

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3.4.25 (continued)
selector shall continuously sample the signal delivered from each of the two antennas in an alternating manner such that the antenna transfer control voltage is s witched to effect transfer of antennas not less than 3 nor more than 6 seconds after the start of the sampling period for a given antenna. The sampling period shall be adjustable between these limits.

3.4.26 Stability - The receiver-transmitter frequency stability shall be as follows:

3.4.26.1 Transmitter Frequency Stability - The frequency of the transmitter shall be maintained within ±0.0087% of the specified channel frequency.

3.4.26.2 Receiver Frequency Stability:

3.4.26.2.1 Local Oscillator Stability - The frequency of the receiver local oscillator shall be maintained within ±0.0087% of its nominal value.

3.4.26.2.2 I-F Stability - The center frequency in the IF passband shall be maintained within ±36 KHz of its nominal value.

3.4.27 Automatic Gain Control (AGC) - The AGC signal shall be generated subsequent to decoding and the AGC bus impedance shall be consistent with the equipment performance requirements. In the A/A mode the equipment shall not lock-on to any reply which is more than 12 db weaker than any other reply.

3.4.28 Blanking Pulses:

3.4.28.1 Internally Generated Blanking Pulse - This receiver-transmitter shall generate a single blanking pulse, with characteristics specified below, for each interrogation pulse pair transmitted and for each single pulse reply transmitted. The blanking pulse shall be capable of suppressing other pulse equipments in the same frequency range, including another equipment identical to the one described by this specification.

(1) Positive polarity.

(2) Not less than 15 nor more than 70 volts amplitude across 500 ± 50 ohms in parallel with 1000 uuf load impedance.

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(3) Not 1ess than 20 volts per microsecond rate of pulse rise (10% to 90% of pulse amplitude).

(4) Not less than 10 volts per microsecond rate of pulse decay (90% to 10% of pulse amplitude).

(5) 25 ± 3 microseconds duration at the 15 volt level.

(6) The 15 volt level on the leading edge shall occur sooner than the 10% amplitude point on the leading edge of the corresponding transmitted reply pulse or first interrogation pulse by an amount not less than 0.8 nor more than 5.0 microseconds.

(7) Any spurious output at the blanking pulse terminal shall not be more than one volt in amplitude.

3.4.28.2 Blanking from External Source - The equipment shall provide for blanking its own receiver during the transmission time of other pulse equipments in the same frequency range when a blanking pulse from an external equipment is applied to the receiver-transmitter. The blanking pulse characteristics will be the same as those defined in 3.4.28.1, except that the blanking pulse duration shall be long enough to effectively bracket the transmission time.

3.4.29 Transmitter Muting - The transmitter shall be automatically muted during channeling in the T/R and A/A modes.

3.4.30 Bearing Circuitry - The equipment shall provide a course deviation output for use with from one to three ID-387/ARN Course Indicators, and an automatic bearing output for display on one to three ID-663/U Indicators in parallel. The bearing circuitry shall be capable of operating on 15 cycle signal alone in the absence of 135 cycle signal. Reduced accuracy will be tolerated in this mode of operation, consistent with 3.4.16.2.1.

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3.4.30.1. Automatic Bearing Circuitry - The automatic bearing output shall be presented to the ID-663/U as combined magnetic heading-bearing signal.. Magnetic heading data will be available from the aircraft compass system. An output accuracy degradation of ±0.5 degree is permissible when the equipment is loaded with an impedance equivalent to three (2) ID-663/U instruments in parallel.

3.4.30.2 Course Deviation Circuitry - The course deviation output shall be presented to the vertical needle of the ID-387/ARN as a deviation from the manually selected course to or from the beacon). The impedance into which it operates shall be maintained at 333 ohms.

3.4.30.2.1 Course Deviation Sensitivity - Course deviation sensitivity shall adjustable so that deviations of from 2.5° to 15° will provide 450 microarmperes into a load impedance of 333 ohlns.

3.4.30.2.2 To-From Indicator - The receiver-transmitter shall provide outputs suitable to operate the To-From Indicator in the ID-387/ARN.

3.4.30.3 Flap Alarm - The flag alarm of the ID-387/ARN shall be actuated and the flag shall disappear when reliable signals above minimum usable are received from the ground station. Upon loss of signal, the flag shall reappear. The flag shall be in view in the absence of either 15 cycle or 135 cycle signals, despite the fact that the automatic bearing indicator may be operating on the 15 cycle signal only. Flag operation is also required when the percentage of modulation of the 15 cycle or 135 cycle signal is below the point where adequate phase stability between the two signals can be maintained.

3.4.31 Distance Circuitry - The distance measuring circuitry shall determine automatically the distance in nautical miles of the equipment from the transponder by measuring the time delay between the transmitted interrogation pulses and the reply pulses.

3.4.31.1 Distance Flag Alarm - The equipment shall be capable of operating (3) indicator flag alarms to show either presence or absence of an adequate signal. The flag shall be in full view during the memory period.

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3.4.32 Outputs:

3.4.32.1 Identify Output - The audio identity tone output shall be adjustable at the control unit to provide a maximum of not less than 100 milliwatts of 1350 cps power into a 600 ohm resistive load during identity tone operation in the REC and T/R modes. The audio power output shall not vary more than 3 db over an RF input signal range of -90 dbm to zero dbm. Accidental shorting or opening of the output circuit external to the equipment shall not cause damage to the equipment or render other functions of the equipment inoperative.

3.4.32.1.2 Identitv Tone Impedance - The control circuit shall provide for identity tone input and output nominal impedance of 600 ohms.

3.4.32.2 Distance Outputs - The Range Mechanical module (Paragraph 3.5.1.4(5) shall provide the following distance outputs for a computer.

3.4.32.2.1 Distance Synchro Transmitters - Three distance synchro transmitters shall be provided, Each synchro shall be positioned in such a manner as to provide distance data of a nature regarding units, tens and hundreds, respectively. A syn - thetic hundreds synchro transmitter is permitted.

3.4.32.3 Bearing Outputs - The Bearing Mechanical module of 3.5.1.4(9) shall provide the following bearing outputs for a computer.

3.4.32.3.1 TACAN Bearing Output - The Bearing Mechanical module shall provide a synchro output of constant phase relationship to the TACAN bearing.

3.4.32.3.2 Bearing Differential Synchro Transmitter Output - A relative bearing output shall be provided by a differential synchro transmitter whose inputs shall be the TACAN bearing angle and the heading reference signal.

3.5 Detail Requirements:

3.5.1 Receiver-Transmitter - RT-1030/ARN-105- The receiver-transmitter shall meet the following requirements.

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3.5.11 Function - The receiver-transmitter shall encode and transmit interrogation pulses; it shall detect and decode bearing, distance and identity data, and convert the data into signals for operation of any combination of:

(1) Three (3) Bearing-Distance - Heading Indicators, ID-663/U, or electrical equivalent, connected in parallel,

(2) Three (3) Course (Deviation) Indicators, ID-38 /ARN or electrical equivalent.

(3) Three (3) Range Indicators, ID-388/APN, vice tlie distance portion of the ID-663/U.

The receiver-transmitter shall also transpond to the interrogation pulses of similarly equipped aircraft. It shall transmit single pulse replies to as many as five such interrogating aircraft provided their signal strengths are within 12 db of each other at the receiver input terminal.

3.5.1.2 Form Factor - The form factor of the RT-1030/ ARN-105 receiver-transmitter shall be as shown in Figure 1.

3.5.1.3 Weight - The weight of the receiver-transmitter shall not exceed 48.0 pounds.

3.5.1.4 Contents - The receiver-transmitter unit shall consist of a base assembly and the following modules. Each module shall conform to the provisions of Specification MIL-E-19600. The classification designations shall be A45Y3. However, the altitude denoted by the digit "3" shall be 70,000 feet.

(1) RF Module - 4-1/2 inches x 6 inches x 12 inches

The RF module shall contain a resistance bridge controlled servo system which drives a turret containing 126 crystals (one for each channel) and which tunes the preselector, frequency multiplier and transmitter. This module shall also contain the preselector, double conversion i-f amplifier, high-voltage power supply, modulator, frequency multiplier, and transmitter" subassemblies.

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3.5.1.4 (1) (continued)

The transmitter power amplifier stages shall be cathode modulated and each cavity shall be individually gasket scaled to maintain rated power at an altitude of 70,000 feet.

(2) Range Decoder Module - 4-1/2 inches x 4-1/2 inches x 2 inches

The Range Decoder module shall contain the coincidence decoder circuit, the low impedance AGC generator, aural identification circuitry and a 4.0457 KHz crystal oscillator.

(3) Range A Module - 4-1/2 inches x 4-1/2 inches x 1-1/2 inches

The Range A module shall contain the modulator trigger and the range gate circuits.

(4) Range D Module - 4-1 /2 inches x 4-1 /2 inches x 2 inches

The Range B module shall contain the early and late gate coincidence amplifiers and the circuits necessary to drive the distance motor control magnetic amplifier.

(5) Range Mechanical Module - 4-1/2 inches x 4-1/2 inches x 2 inches

The Range Mechanical module shall contain the motors, tachometers, potentiometers, synchro transmitters and resolvers necessary to provide the ranging circuit with phase shifted or varying potential signals necessary to measure distance and to provide outputs to drive remote distance indicators.

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(6) Bearing Decoder Module - 4-1/2 inches x 4-1/2 inches x 2 inches

The Bearing Decoder module shall contain the burst eliminator, peak riding detector and the 15 andl 135 cycle reference blocking oscillators. The module furnishes 15- and 135- cycle bearing information to the bearing circuits.

(7) Bearing A Module - 4-1/2 inches x 4-1/2 inches x 2 inches

The Bearing A module shall contain the 15-cps filter circuits, a sine-cosine card phase splitter to feed the 15- cps phase shifter, 15- and 135- cps phase comparison circuits, 15- and 135- cps switching network and motor circuits.

(8) Bearing B Module - 4-1/2 inches x 4-1/2 inches x 2 inches

The Bearing B module shall contain the 135 cps filter circuits, 40-degree gate forming circuits and search-track control circuits.

(9) Bearing Mechanical Module - 4-1/2 inches x 4-1/2 inches x 2 inches

The Bearing Mechanical Module shall contain the necessary resolvers, potentiometers, synchro transmitters, tachometers and motors to furnish the 15- and 135-cps phase shifts and to drive remote bearing indicators.

(10) Magnetic Amplifier and Phase Detector Module - 4-1/2 inches x 4-1/2 inches x 1-1/2 inches

The Magnetic Amplifier and Phase Detector module shall contain two magnetic amplifiers, one each for the bearing and distance motor drive circuits. It shall also contain a phase

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(10) (continued)
detector which compares the phase of a signal from tile bearing mechanical module with the phase 01 a manually selected signal and provides an output proportional 10 deviation from the selected bearing.

(11) Power Supply Module - 4-1/2 inches x 4-1/2 inches x 4 inches

The Power Supply module shall contain Circuitry for generating the power required by the receiver-transmitter, except the high voltage power required by the r-f power amplifiers. A nominal 90 second time delay circuit shall also be included in this module.

(12) Air-to-Air Range Module - 4-1/2 inches x 4-1/2 inches x 1 inch

The Air-to-Air Range module shall contain the circuits necessary in the air-to-air mode to accept an interrogation, provide the required system delay and trigger the modulator to reply to the interrogation. Also, an auxiliary AGC circuit, a range compensation network to the Range A module and a muting control voltage to the modulator in the RF module shall be provided.

(13) Antenna Selector Module - 4-1/2 inches x 4-1/2 inches x 1 inch

The Antenna Selector moDule shall contain circuits to provide an output to an external RF antenna selector switch. The limited, decoded TACAN signal from the Range Decoder module shall be used to distinguish between usable and unusable signals, and the antenna selector module circuitry shall make selections based on this output.

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3.5.1.4.1 Contents - Receiver-transmitter RT-1030/ ARN-105 shall, in addition to modules of paragraph 3.5.1.4, contain the following:

(1) A basic module-supporting double-walled structure, containing a harness and acting as a cooling plenum.

A step-down transformer for 115/26 VAC instrument phase.

A tone transformer to isolate the ground on the identity tone output of the Range Decoder Module.

(4) Wiring and electronic components performing the electrical function of Mount MT-1729/ ARN-52(W), MT-1729A/ARN-52(V), MT-3680A/ ARN-52(V) or MT-3952/ARN-2(V).

(5) In addition to (1) through (4) above, components necessary to achieve interchangeability in form, fit and function (with additional capabilities) with the RT-471/ARN-65, and requiring no modifications to aircraft wiring or instrumentation.

3.5.1.4.2 Module Reliability in Mean-Time-Between-Failures (MTBF) - The modules shall have the following listed mean (operating time between failures when tested and accepted as outlined under the requirements of 4.4.3.2.

Module	MTBF
Range A	3000 hours
Range B	6500 hours
Bearing A	5000 hours
Bearing B	5000 hours
Range Decoder	2500 hours

..... cont'd.

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Module	MTBF
Bearing Decoder	5090 hours
Power Supply	5000 hours
RF Module	1000 hours
Range Mechanical	7200 hours
Bearing Mechanical	7200 hours
Msg. Amp.	2000 hours
Air-to-Air	2000 hours
Antenna Selector	2000 hours

3.5.1.5 Controls - A screwdriver adjustment, marked COURSE WIDTH ADJ. shall be provided on the front panel of the receiver-transmitter unit for adjusting the course deviation, from center to full left or right, 10 values between 2.5 and 15.0 degrees.

3.5.1.6 Electrical Connections - Electrical Connections to external circuits shall be as shown in Figure 3.

3.5.2 Control C-2010/ARN-52(V) - The control shall meet the applicable requirements of Specification MIL-C-6781 for Type I control panel. The lighting of the plastic panel shall conform to MlL-P-7788, Class 1R, Type III with a lamp voltage of 28 volts.

3.5.2.1 Function - The control unit shall perform the following functions:

(1) It shall provide for applying and turning off power to the receiver-transmitter; it shall also provide for manual selection of the REC, T/R and A/A modes of operation.

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3.5.2.1 (continued)

(2) It shall provide for manual selection of and one of 126 channels assigned to the receiver-transmitter.

(3) It shall provide for manual control of the volume of the aural tone of the beacon identification.

3.5.2.2 Form Factor - The form factor 01 Control C-2010/ ARN-52(I') shall be as shown in Figure 2.

3.5.2.3 Weight - The weight of the control unit shall not exceed 2 pounds.

3.5.2.4 Contents - The control unit shall contain the following:

F'unction Selector Switch
Channel Selector Switch
Ideniity Tone Output Control
Panel. Lights
Channel Servo Bridge Circuits
Connector MS3102A-20-27S

3.5.2.5 Controls - The Control unit shall provide the following controls:

3.5.2.5.1 Function Selector - The function selector shall be a four position rotary switch to provide selection of the operating functions which shall be marked OFF, REC, T/R, and A/A, respectively in a clockwise order. The selector switch shall control the receiver-transmitter as follows:

(1) In the OFF position, all primary power to the receiver-transmitter shall be off.

(2) In the REC position, the receiver-transmitter shall be energized, allowing the receiver portion of the unit to operate for bearing and identity tone functions on any channel.

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3.5.2.5.1 (continued}

(3) In the T/R position, the transmittcr portion of the unit will also be energizcd and in addition to bearing and identity tone functions will allow the transmission of interrogation pulses to the ground beacon and receive the distance response pulse from the beacon as distance information.

(4) In the A/A position, the receiver-transmitter shall be energized and perform as a transponder for distance information between cooperating a.ircrafts. The bearing circuits shall he inactive in this mode.

3.5.2.5.2 Channel Selector - The channel selector shall consist of two rotary switches on concentric shafts. The switch on the inner shaft shall contain 10 positions and shall effect a digital readout of zero 10 nine respectively as the switch is moved clockwise through its range. The switch on the outer shaft shall contain 13 positions and shall effect a digital readout of onc to twelve as the switch is moved clockwise from its second position to its thirteenth position respectively. There shall be no readout in the most counterclockwise position. The digit displayed by the inner control shall appear immediately below tile number displayed by the outer control. When the number displayed by the outer control contains two digits, the number shall be displayed vertically so that the second digit is immediately below the first digit. The channel selector shall be marked CHAN immediately above the switch knobs.

3.5.2.5.3 Identity Tone Output Control - The identity tone output level shall be controlled by a potentiometer assembly which shall provide continuous variation of the audio output level, approximately linearly in db per degree of control rotation, over a range of attenuation from 60 db to a minimum of not more than 0.5 db. A clockwise rotation shall effect an increase in output level. The control shall be marked VOL between the forward and rear portions of a clockwise arrow broken in the middle to accommodate the letters.

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3.5.2.6 Electrical Connections - The electrical connections to external circuits shall be as shown in Figure 3.

3.5.2.7 Panel Lights - Panel lights for the C-2010/ARN-52(V) shall utilize type MS25237-327 lamps and MS25010A11A lampholders. These lamps are rated 28 volts and tile lampholders have a red lens.

3.5.3 Control C-2010A/ARN-52(V) - The control shall mmet the applicable requirements of Specification MIL-C-6781 for Type I control panel. The lighting of the plastic panel shall conform to MIL-P-7788, Class 1R, Type III with a lamp voltage of 28 volts.

3.5.3.1 Functions - The control unit shall perform the following functions.

(1) It shall provide applying and turning off power to the receiver-transmitter; it shall also provide for manual selection of the REC, T/R, and A/A mocles of operation.

(2) It shall provide for manual selection of any one of 126 channels assigned to the receiver-transmittcr。 It shall at the same time provide digital outputs corresponding to the channel selected,

(3) It shall provide for manual control of the volume of the aural tone of the beacon identification.

3.5.3.2 Form Factor - The for m factor of Control C-2010A/ARN-52(V) shall be as shown in Figure 4.

3.5.3.3 Weight - The weight of the Control unit shall not exceed 2.5 pounds.

3.5.3.4 Contents - The control unit shall contain the following:

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Function Selector Switch
Channel Selector Switch including digital switching
Identity Tone Output Control
Panel Lights
Channel Servo Bridge Circuit
Connectors MS-3102A-20-27S and
MS-3102R-20-33S

3.5.3.5 Controls - The control unit shall provide the following controls:

3.5.3.5.1 Function Selector - The function selector shall be a four position rotary switch to provide selection of the operating functions which shall be marked OFF, REC, T/R and A/A respectively in a clockwise order. The selector switch shall control the receiver-transmitier as follows:

(1) In the OFF position, all primary power to the receiver-transmiter shall be off.

(2) In the REC position, the receiver-transmitter shall be (energized, allowing the receiver portion of the unit to operate for bearing and identity tone functions on any channel.

(3) In the T/R position, the transmitter portion of the unit will also be energized and in addition to the bearing and identity tone functions, will allow the trasmission of interrogation pulses to the ground beacon and receive the distance response pulse from the beacon as distance information.

(4) In the A/A position, the receiver-transmitter shall be energized and perform as a transponder for distance information between cooperating aircrafts. The bearing circuits shall be inactive in this mode.

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3.5.3.5.2 Channel Selector - The channel selector shall consist of two rotary switches on concentric shafts. The switch on the inner shaft shall contain 10 positions and shall effect a digital readout of zero to nine respectively as the switch is moved clockwise through its range. The switches on the outer shaft shall contain 13 positions and shall effect a (digital readout of one to twelve as the switch is moved clockwise from its second position to its thirteenth position respectively. There shall be no readout in the most counter-clocktvise position. The digit displayed by the inner control shall appear immediately below the number displayed by the outer control. When the number is displayed by the outer control contains two digits, the number shall be displayed vertically so that the second digit is immediately below the first digit. The channel selector shall be marked CHAN immediately above the switch knobs. In addition to providing for the selection of channels, the channel selector switch shall provide digitial switching corresponding to the channel selected by means of a seven (7) bit natural binary output. Each binary output lead shall, in the mark condition, provide connection to a binary common lead with a maximum series resistance of fifty (50) ohms. An external voltage source may be fed into the common lead for distribution to the binary outputs. Each binary output lead shall, in the space condition, be open circuited with respect to the binary common lead. In either condition, however, a nominal source impedance of 5000 ohms shall be provided when measured with respect to the binary circuit ground.

3.5.3.5.3 Identity Tone Output Control - The identity tone output level shall be controlled by a potentiometer assembly which shall provide continuous variation of the audio output level, approximately linearly in db per degree of control rotation, over a range of attenuation from 60 db to a minimum of not more than 0.5 db. A clockwise rotation shall effect an increase in output level. The control shall be marked VOL between the forward and rear portions of a clockwise arrow broken in the middle to accommodate the letters.

3.5.3.6 Electrical Connections - The electrical connections of the C-2010A/ARN-52(V) control unit shall comply with Figure 4.

3.5.3.7 Control C-3928(P)/ARN-52(V) - The control unit C-3928(P)/ARN-52(V) will be identical to the C-2010/ARN-52(V) control unit (para. 3.5.2) except as follows:

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3.5.3.7.1 Nameplate - The nameplate shun designate control C-3928(P)/ARN-52(V) instead of C-2010/ARN-52(V).

3.5.3.7.2 Panel Lights - The panel lights for the C-3928(P)/ ARN-52(V) shall utilize type MS25237-328 lamps and MS25010A11A lampholders. These lamps are rated at 6 volts and the lampholders have a red lens.

3.5.3.8 Control C-7893(P)/ARN-52(V) - The control unit C-7893(P)/ARN-52(V) will be identical to the C-2010/ARN-52(V) control unit (para. 3.5.2) except as follows:

3.5.38.1 Nameplate - The nameplate shall designate control C-7893(P)/ARN-52(V) instead of C-2010/ARN-54(V).

3.5.3.8.2 Panel Lights - The panel lights for the C-7893(P)/ ARN-52(V) shall utilize type MS25237-328 lamps and MS25010A12A lampholders. These lamps are rated 6 volts and the lampholders have a clear lens.

3.5.3.9 Control C-7712/ARN-52(V) - The control unit C-7712/ARN-52(V) will be identical to the C-2010/ARN-52(V) control unit (para. 3.5.2) except as follows:

3.5.3.9.1 Nameplate - The nameplate shall designate control C-7712/ARN-52(V) instead of C-2010/ARN-52(V).

3.5.3.9.2 Panel Lights - The panel lights for the C-7712/ ARN-52(V) shall utilize type MS90452-7153 white lamps embedded in the panel. These lamps are rated 5 volts.

3.5.4 Mounting MT-2091/ARN-65(V) - A vibration and shockabsorbing mounting MT-2091/ARN-65(V) shall be provided for the RT-1030/ARN-105 receiver-transmitter with a form factor as shown on Figure 5. Its loaded height shall not exceed 2.25 inches.

3.5.4.1 Function - The mounting shall provide for vibration shock absorption in accordance with MIL-C-172.

3.5.4.2 Weight - The weight of the MT-2091/ARN-65(V) mounting shall not exceed 2.25 pounds.

3.5.4.3 Contents - The mounting shall contain the following:

Shock Isolators (5)
Ground Straps (2)
Guide Pins (2)

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his oven facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine

(1) Preproduction (First Article) Tests
(2) Initial Production Tests
(3) Acceptance Tests
(4) Life Tests

4.2 Preproduction (First Article) Tests - Preproduction tests shall be conducted by the contractor on an equipment representative of the production equipments to bc supplied under the contract, Preproduction tests shall bc accomplished under the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a government representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring activity for review and approval.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification, other applicable spccifications and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422, modified to meet the requirements of paragraphs 3.3.10.1, 3.3.10.2 and 3.3.10.3.

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4.2.2.1 Acceleration Tests - Acceleration Tests shall be conducted in accordance with Procedure III, MIL-E-5272. Functional tests shall be performed in accordance with the approved preproduction test procedure (Part III).

4.2.2.2 Acoustical Noise Tcsts - Accoustical Noise tests shall be concluc ied in accordance with J'letilod 515, MIL-STD-810A except the Tolerances of Figure 515-1 MIL-STD-810A, with an overall sound pressure level of 145 db (re $2x10^{-4}$ dy/cm^2) shall apply. Functional tests shall be performed in accordance with the approved preproduction test procedure (Part III).

4.2.2.3 Interference Tests - Interference tests shall be conducted in accordance with Specification MIL-T-6181.

4.2.3 Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance cnd quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activtity after the equipment has successfully passed all individual tests. No other tests shall be conducted on equipment prior to starting the Initial Production Tests. The preproduction sample shall not be selected for this test.

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4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipmen is equivalent to the previously approved preproduction sample in design, construction, workmanship, performances and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisifactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction fo the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipemnt previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all limited life or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all sample's and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

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(1) Individal Tests
(2) Sampling Tests
(3) Reliability Assurance Tests
(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product
(2) Operational Test
(3) Manufacturing Run in Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test. Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run in Test - Each equipment shall be operated under the conditions specified herein for a period of 10 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which cm be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature - Ambient - Room
Humidity - Ambient - Room
Vibration - Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3 g's.

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The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 10-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 10-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 10-hour period specified above may be composed of two 5-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government inspector in accordance with the following:

Quaniity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	1#
Next 50	1
AText 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

#One out of first ten need not be selected and tested if initial production tests are conducted. Sampling Tests are not required when Reliability Assurance Tests are conducted.

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4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operation test at ambient roonl Conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Opcrational test at certain environmental conditions, The conditions may vary for each equipmcnt tested and should be based on results of the preproduction, production and special tests.

(3) Manufacturing run in test specified in paragraph 4.4.1.3 except that the test duration shall bc 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Production Acceptance (Sampling) Phase shall be conducted. Classification of failure shall be in accordance with MIL-STD-781 and AR-34.

4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level E shall be used. The Accept-Reject Criteria for Test Plan H shall be used.

4.4.3.2 Reliability Production Acceptance Phase Tests - The equipment throughout production shall be tested as outlined in MIL-STD-781 (as modified herein) under the section entitled "Production Acceptance (Sampling) Phase of Production Reliability Tests", except that all equipments produced shall be tested. Test Level E of MIL-STD-781 shall be used.

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4.4.3.2.1 All Equipment Test - Each equipment produced shall be tested for 50 hours. Prior to the 50 hour test on each equipment, a burn-in period may be used at the option of the contractor. If the burn-in period is to be used the details thereof must be included in the approved test procedures. To determine whether the MTBF is being met at any time during the contract the operating test hours and failures thereon (not counting burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject ine of Test Plan II of MIL-STD-781. (Extend the line as necessary to accommodate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attention: Avionics Division. At any time that the current totals of test hours and test failures plotted on Test Plan II curves show a reject situation, the procuring activity reserves the right to stop the acnotified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision.

4.4.4 Special Tests - Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of ihe equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.1.1.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

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4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, the proposed corrective actioh intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the life tests. The test shall be of 300 hours duration and shall be conductcd on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test samples shall be selected by the Governmcnt inspector in accordance with the following: (Equipments which have successfully passed tile initial Production Test, Sampling: Tests, Reliability Tests, or Special Tests may be selected for Life Tests.) (Test time accumulated during the reliability Test may be counted toward the Life Test provided the entire 300 hours is accumulated on a single equipment and equipment selected is in accordance with the table below.)

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Quality of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next. 300	1
	1 for each additional 500 or fraction thereof.

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions.

Temperature	Normal Room
Altitude	Normal ground (0 - 5000 ft.)
Humidity	Room ambient
A. C. Voltage	115 ± 5 volts (at lowest applicable frecluency)
D. C. Voltage	2'7.0 ± 2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check -- At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencics or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

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(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacture

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure.

4.5.4.1 Failure Report - In the event of a failure, the Government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test, In the report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance test and life tests shall bc prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification of the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to the initial production, acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equjpment for acceptance.

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4.8 Premission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specifications MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in Specification MIL-E-17555, the method of preservationn for level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Intended Use - The equipment provides distance and bearing information to an aircraft pilot. It is used in fixed and rotary-wing aircraft, and the information is referenced to a land or carrier-based beacon, or to other similarly equipped aircraft.

6.2 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification

(2) Selection of applicable levels of packaging and packing (see 5.1)

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6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable or subsidiary specificcations are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specificiation, or from subsidiary specifications where applicable, shall be specifically approved in writintg by the procuring activity

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification me objectives which shall. be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 Type Designations - The type designation may be modified b.y the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The correct type number shall be used in nameplates, shipping recol.ds and instruction books, as applicable.

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6.6 Revisions - In specification revisions and superseding amendments an asterisk (*) preceing a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.7 Associated Equipment - The equipment shall operate with the following associated equipment, or their equivalents. which shall not be supplied as part of the AN/ARN-105. The equipment will provide necessary inputs for driving up to three ID-663/U, or AQU-2/A, or AQU-4/A or ID-1013/A or any equivalent combination thereof. In addition, it will provide outputs for instrument combinations indicated in paragraph 3.5.1.1.

(1)	Indicator, Bearing-Distance-Heading ID-663/U	(MIL-22075)
(2)	Indicator, Course ID-387/ARN	(MIL-19229)
(3)	Indicalor, Course ID-250A/ARN	(MIL-C-58Z4)
(4)	Indicator, Range ID-388/APN	(MIL-I-18955)
(5)	Switch, Radio Frequency Transmission Line SA-521/A	(MIL-S-25879)
(6)	Antenna AT-741/A (or electrical equivalent Electonic Resources, Inc. PN 01-23-03721	(MIL-A-25708)
(7)	Horizontal Situation Indicator ID-1013/A	(MIL-I-23366)

. ..cont'd..

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(8) Horizontal Situation Indicator AQU-47A (MIL-I-27048)

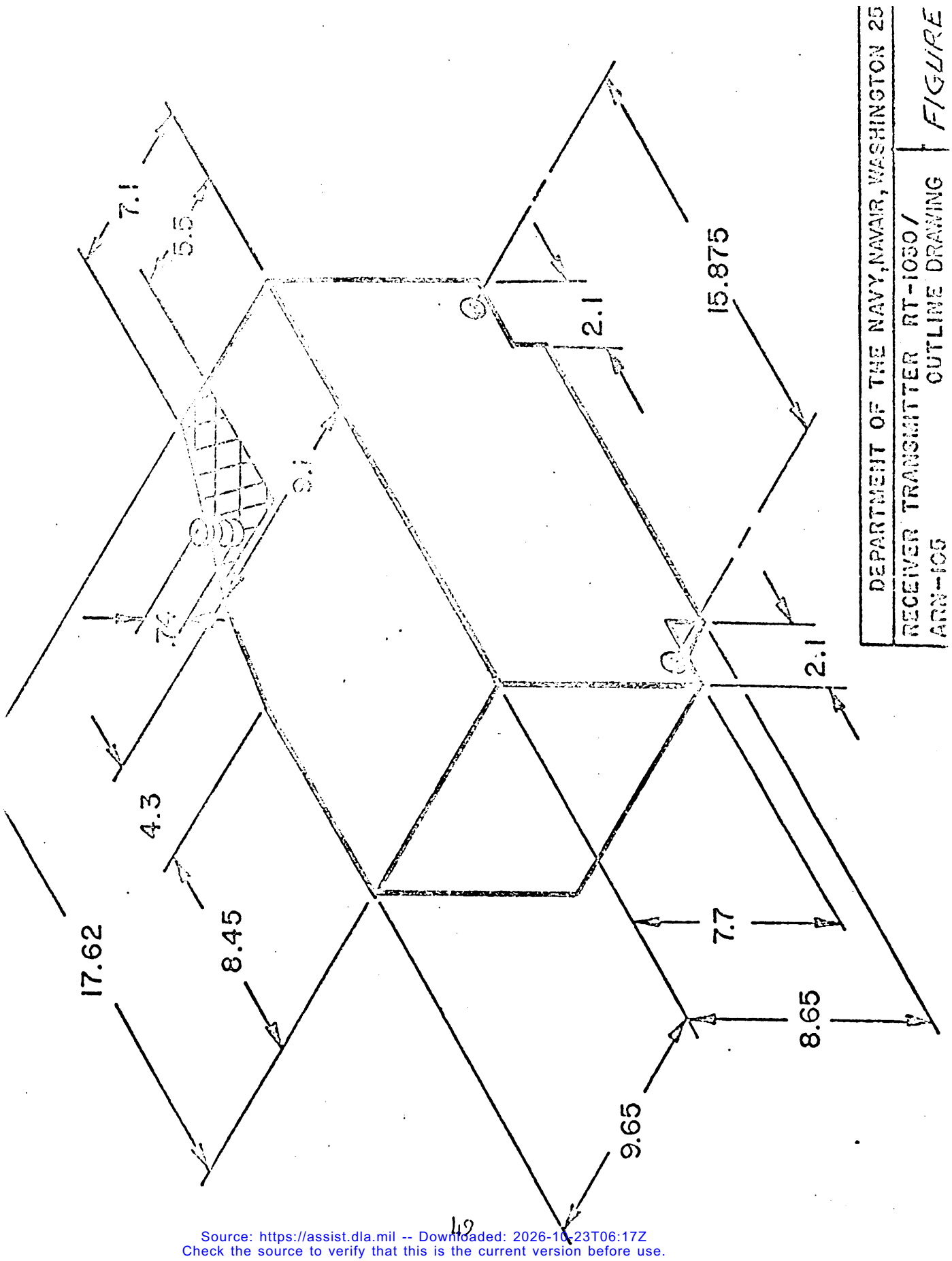
(9) Horizontal Situation Indicator AQU-2/A (MIL-H-26689)

(10) Indicator, Bearing-Distance Heading ID-526A (MIL-I-25990)

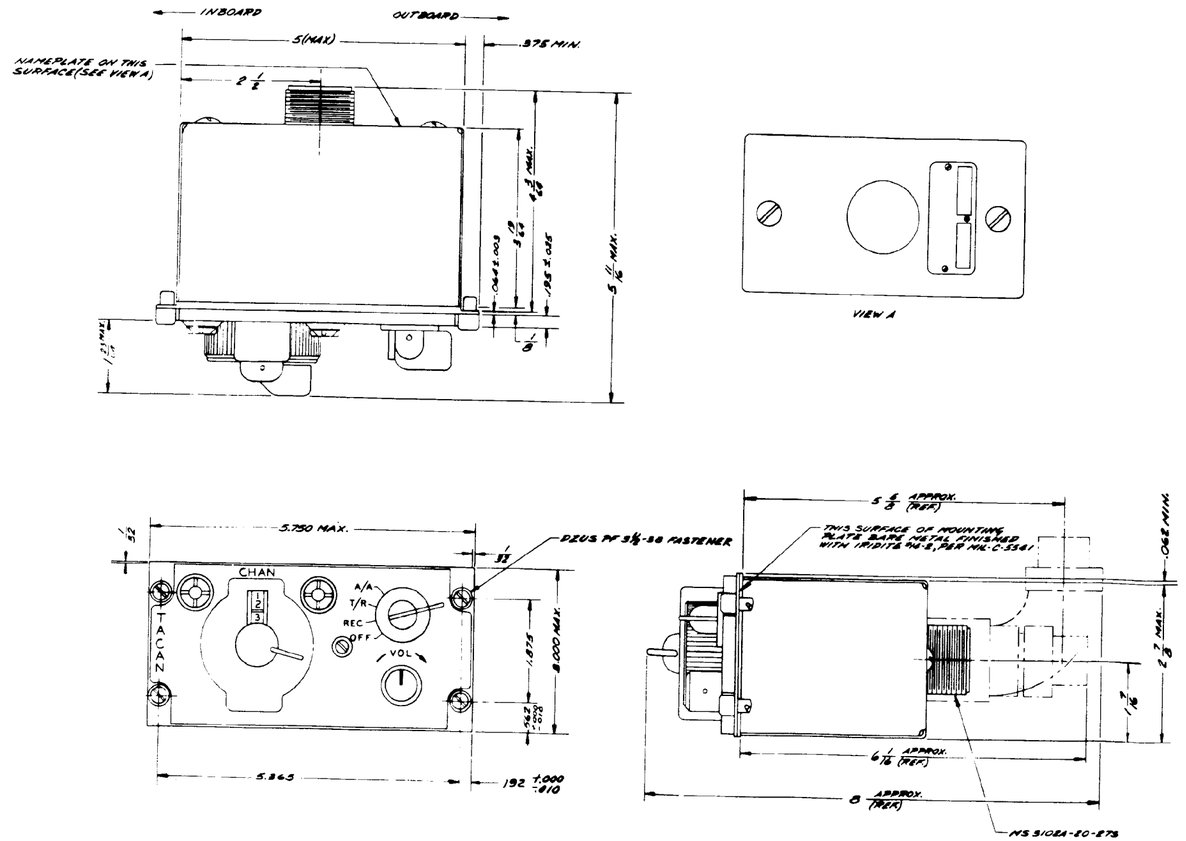
6.8 This specification is under the cognizance of AIR-53373A.

Project Number 5826-N108

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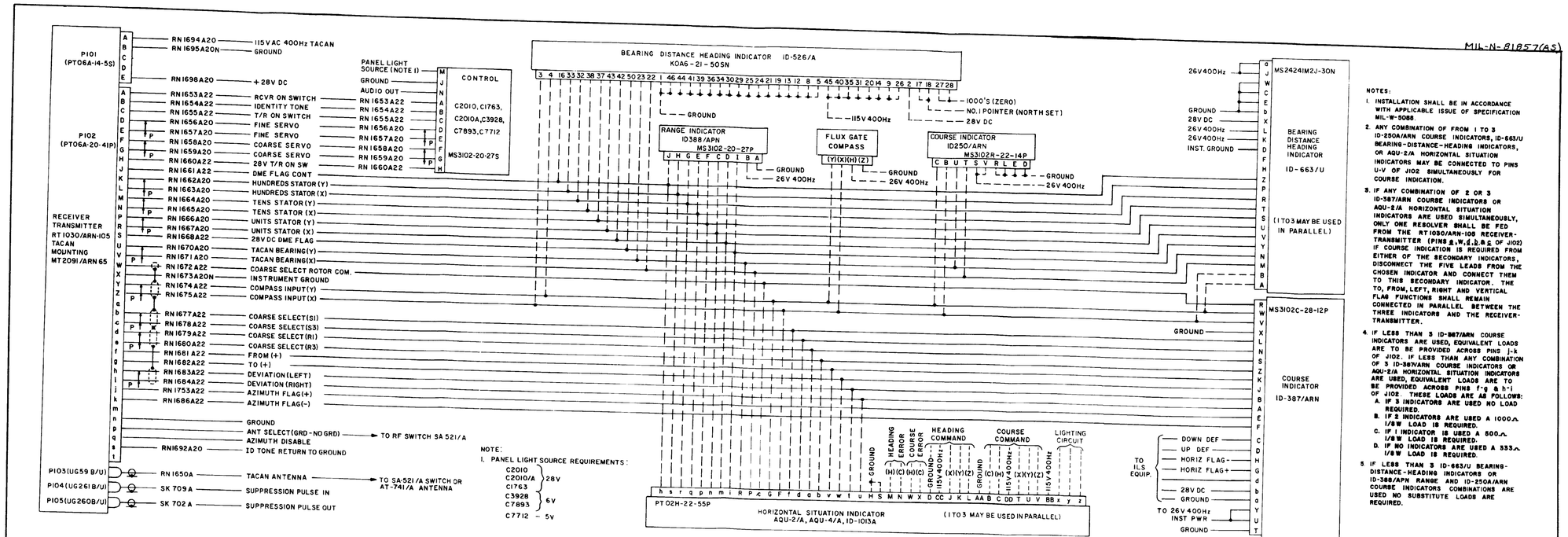


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CONTROL BOX C-2010/ARN-52(V) OUTLINE DRAWING	FIG 2

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NOTES:

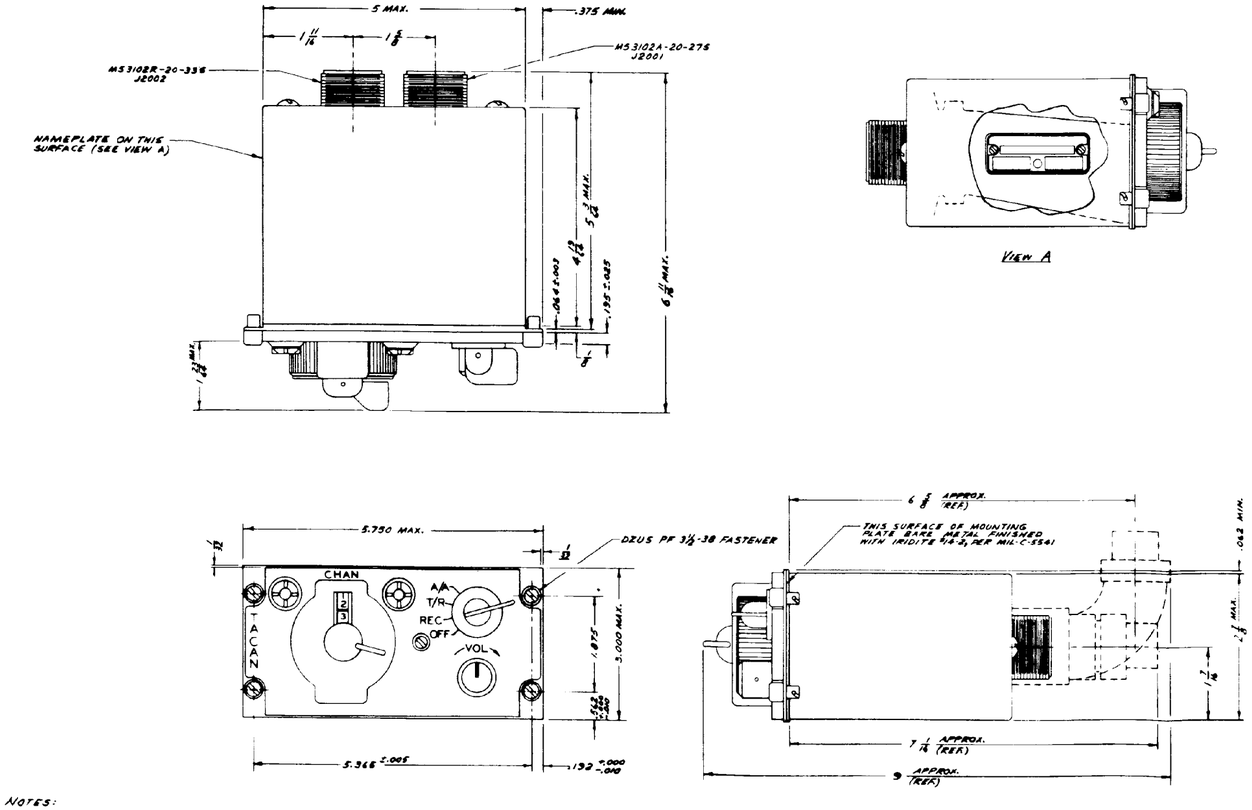
1. INSTALLATION SHALL BE IN ACCORDANCE WITH APPLICABLE ISSUE OF SPECIFICATION MIL-W-5088.
2. ANY COMBINATION OF FROM 1 TO 3 ID-250A/ARN COURSE INDICATORS, ID-663/U BEARING-DISTANCE-HEADING INDICATORS, OR AQU-2/A HORIZONTAL SITUATION INDICATORS MAY BE CONNECTED TO PINS U-V OF J102 SIMULTANEOUSLY FOR COURSE INDICATION.
3. IF ANY COMBINATION OF 2 OR 3 ID-387/ARN COURSE INDICATORS OR AQU-2/A HORIZONTAL SITUATION INDICATORS ARE USED SIMULTANEOUSLY, ONLY ONE RESOLVER SHALL BE FED FROM THE RT1030/ARN-105 RECEIVER-TRANSMITTER (PINS a, w, d, b & c OF J102) IF COURSE INDICATION IS REQUIRED FROM EITHER OF THE SECONDARY INDICATORS, DISCONNECT THE FIVE LEADS FROM THE CHOSEN INDICATOR AND CONNECT THEM TO THIS SECONDARY INDICATOR. THE TO, FROM, LEFT, RIGHT AND VERTICAL FLAG FUNCTIONS SHALL REMAIN CONNECTED IN PARALLEL BETWEEN THE THREE INDICATORS AND THE RECEIVER-TRANSMITTER.
4. IF LESS THAN 3 ID-387/ARN COURSE INDICATORS ARE USED, EQUIVALENT LOADS ARE TO BE PROVIDED ACROSS PINS J-k OF J102. IF LESS THAN ANY COMBINATION OF 3 ID-387/ARN COURSE INDICATORS OR AQU-2/A HORIZONTAL SITUATION INDICATORS ARE USED, EQUIVALENT LOADS ARE TO BE PROVIDED ACROSS PINS f & h-i OF J102. THESE LOADS ARE AS FOLLOWS:
   A. IF 3 INDICATORS ARE USED NO LOAD REQUIRED.
   B. IF 2 INDICATORS ARE USED A 1000Ω 1/8W LOAD IS REQUIRED.
   C. IF 1 INDICATOR IS USED A 500Ω 1/8W LOAD IS REQUIRED.
   D. IF NO INDICATORS ARE USED A 333Ω 1/8W LOAD IS REQUIRED.
5. IF LESS THAN 3 ID-663/U BEARING-DISTANCE-HEADING INDICATORS OR ID-388/APN RANGE AND ID-250A/ARN COURSE INDICATORS COMBINATIONS ARE USED NO SUBSTITUTE LOADS ARE REQUIRED.

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EXTERNAL WIRING DIAGRAM
NAVIGATIONAL SET AN/ARN-105

FIG 3

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NOTES:
1. THE WEIGHT OF THIS UNIT IS LBS.
2. INTERPRET THIS DWG. IN ACCORDANCE WITH STANDARDS PRESCRIBED BY MIL-D-7032X.

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CONTROL BOX C-2010A/ARN-52 (V) OUTLINE DRAWING	FIG 4

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